

## K-Fold Cross-Validation

- If we had enough data, we would set aside a validation set and use it to assess the performance of our prediction model.

$\hookrightarrow$  *Data are often scarce: usually not possible.*

- To bypass this problem,  $k$ -fold CV uses part of the data to fit the model, and a different one to test it.
- For  $K = 5$ , we would have 5 roughly equal sized parts s.t.:

|                       |                       |                       |                       |                       |
|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| <del>Validation</del> | Train                 | Train                 | Train                 | Train                 |
| Train                 | <del>Validation</del> | Train                 | Train                 | Train                 |
| Train                 | Train                 | <del>Validation</del> | Train                 | Train                 |
| Train                 | Train                 | Train                 | <del>Validation</del> | Train                 |
| Train                 | Train                 | Train                 | Train                 | <del>Validation</del> |

- Let  $\hat{f}^{-k}(\mathbf{x})$  the fitted function computed with the  $k$ -th part of the data removed.
- Let the  $K$  parts be  $C_1, \dots, C_K$  with respective number of observations

$$N_k = N/K \quad \left( \text{so that } \sum_{k=1}^K N_k = K \cdot \frac{N}{K} = N \right).$$

- Let  $\kappa : \{1, \dots, N\} \mapsto \{1, \dots, K\}$  indicating the partition to which  $i$  is allocated by the randomization.

### The CV criterion

- The CV estimate of prediction error is:

$$CV(\hat{f}) = \frac{1}{N} \sum_{i=1}^N L(y_i, \hat{f}^{-\kappa(i)}(\mathbf{x}_i))$$

- Typically, researchers pick  $K = 5$  or 10.
- When  $K = N$ , we have  $\kappa(i) = i$ . This setup is called *Leave-One-Out CV (LOOCV)*.

- Given a set of models  $f(\mathbf{x}, \alpha)$ , where  $\alpha$  is a tuning parameter, we have:

$$CV(\hat{f}, \alpha) = \frac{1}{N} \sum_{i=1}^N L(y_i, \hat{f}^{-\kappa(i)}(\mathbf{x}_i, \alpha))$$

The goal is then to find  $\alpha$  minimizing  $CV(\hat{f}, \alpha)$ .

- The point is that LOOCV ( $K = N$ ) is an average of highly correlated estimates with low bias (as the training sets are very similar) and will therefore have high variance. Because it also requires  $N$  applications of the learning method, it is subject to a considerable computational burden.
- Again, based on the bias-variance tradeoff, with  $K = 5$ , CV has lower variance, but bias can become a problem depending on the performance of the learning method as a function of training set size.

## LOOCV for linear models

- For linear models, it turns out that the computational burden of LOOCV is not a problem. Indeed, under a squared loss:

$$CV(\hat{f}) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{f}^{-i}(\mathbf{x}_i))^2$$

- For a linear regression model  $Y = X\beta + \varepsilon$  we get:

$$\hat{\beta} = (X'X)^{-1}X'Y \Rightarrow \hat{Y} = X(X'X)^{-1}X'Y = P_X Y.$$

- We then get that  $\hat{f}^{-i} = \mathbf{x}'_i \hat{\beta}_{[i]}$  where:

$$\hat{\beta}_{[i]} = \left( X'_{[i]} X_{[i]} \right)^{-1} X'_{[i]} Y_{[i]}$$

is using the data without the  $i$ -th observation.

- Specifically:

$$X_{[i]} = \begin{bmatrix} \mathbf{x}'_1 \\ \vdots \\ \mathbf{x}'_i \text{ (removed)} \\ \vdots \\ \mathbf{x}'_N \end{bmatrix}, \quad \text{so that:}$$

$$X'X = \begin{bmatrix} \mathbf{x}_1 & \cdots & \mathbf{x}_i & \cdots & \mathbf{x}_N \end{bmatrix}_{(p \times N)} \begin{bmatrix} \mathbf{x}'_1 \\ \vdots \\ \mathbf{x}'_i \\ \vdots \\ \mathbf{x}'_N \end{bmatrix}_{(N \times p)}$$

$$X'X = \mathbf{x}_1 \mathbf{x}'_1 + \cdots + \mathbf{x}_i \mathbf{x}'_i + \cdots + \mathbf{x}_N \mathbf{x}'_N = \sum_{i=1}^N \mathbf{x}_i \mathbf{x}'_i$$

$$\Rightarrow X'_{[i]} X_{[i]} = X'X - \mathbf{x}_i \mathbf{x}'_i.$$

## The leverage shortcut

- For the hat matrix, its  $i$ -th value of the diagonal is:

$$h_i = \mathbf{x}_i'(X'X)^{-1}\mathbf{x}_i$$

- Then:

$$\begin{aligned} \left(X'_{[i]}X_{[i]}\right)^{-1} &= (X'X - \mathbf{x}_i\mathbf{x}_i')^{-1} \\ &= (X'X)^{-1} + \frac{(X'X)^{-1}\mathbf{x}_i\mathbf{x}_i'(X'X)^{-1}}{1 - \mathbf{x}_i'(X'X)^{-1}\mathbf{x}_i} \end{aligned}$$

(by applying the Sherman–Morrison formula)

$$= (X'X)^{-1} + \frac{(X'X)^{-1}\mathbf{x}_i\mathbf{x}_i'(X'X)^{-1}}{1 - h_i}$$

- Similarly,  $X'_{[i]}Y_{[i]} = X'Y - \mathbf{x}_iy_i$ , so that:

$$\begin{aligned} \hat{\beta}_{[i]} &= \left[ (X'X)^{-1} + \frac{(X'X)^{-1}\mathbf{x}_i\mathbf{x}_i'(X'X)^{-1}}{1 - h_i} \right] (X'Y - \mathbf{x}_iy_i) \\ &= \hat{\beta} - (X'X)^{-1}\mathbf{x}_iy_i + \frac{(X'X)^{-1}\mathbf{x}_i\mathbf{x}_i'(X'X)^{-1}}{1 - h_i} (X'Y - \mathbf{x}_iy_i) \\ &= \hat{\beta} - (X'X)^{-1}\mathbf{x}_iy_i + \frac{(X'X)^{-1}\mathbf{x}_i}{1 - h_i} \mathbf{x}_i'\hat{\beta} - \frac{(X'X)^{-1}\mathbf{x}_i}{1 - h_i} h_i y_i \\ &= \hat{\beta} - \frac{(X'X)^{-1}\mathbf{x}_i}{1 - h_i} \left[ y_i(1 - h_i) - \mathbf{x}_i'\hat{\beta} + h_i y_i \right] \\ &= \hat{\beta} - \frac{(X'X)^{-1}\mathbf{x}_i}{1 - h_i} \left( y_i - \mathbf{x}_i'\hat{\beta} \right) \end{aligned}$$

- Hence:

$$\begin{aligned} y_i - \hat{f}^{-i}(\mathbf{x}_i) &= y_i - \mathbf{x}_i'\hat{\beta}_{[i]} \\ &= y_i - \mathbf{x}_i' \left[ \hat{\beta} - \frac{(X'X)^{-1}\mathbf{x}_i}{1 - h_i} \left( y_i - \mathbf{x}_i'\hat{\beta} \right) \right] \\ &= y_i - \mathbf{x}_i'\hat{\beta} + \frac{h_i}{1 - h_i} \left( y_i - \mathbf{x}_i'\hat{\beta} \right) \\ &= \left( y_i - \mathbf{x}_i'\hat{\beta} \right) \left[ 1 + \frac{h_i}{1 - h_i} \right] = \frac{y_i - \mathbf{x}_i'\hat{\beta}}{1 - h_i} \end{aligned}$$

$$\Rightarrow y_i - \hat{f}^{-i}(\mathbf{x}_i) = \frac{y_i - \hat{f}(\mathbf{x}_i)}{1 - h_i}, \quad \text{so that:}$$

$$CV(\hat{f}) = \frac{1}{N} \sum_{i=1}^N \left( \frac{y_i - \hat{f}(\mathbf{x}_i)}{1 - h_i} \right)^2$$

## Generalized Cross-Validation

- For a more general setup where  $\hat{f}(\mathbf{x}_i)$  may not be a linear index, we have  $\hat{\mathbf{y}} = S\mathbf{y}$  and  $\sim_i \equiv h_i$  the diagonal elements of  $S$ . Sometimes, the trace can be more easily computed than its diagonal elements. Using the relation we derived previously:
- The average of the diagonal elements is:

$$\frac{1}{N} \sum_{i=1}^N s_i = \frac{\text{Tr}(S)}{N}, \quad \text{so that we approximate:}$$

$$s_i \simeq \frac{\text{Tr}(S)}{N}$$

and approximate CV as:

$$GCV(\hat{f}) = \frac{1}{N} \sum_{i=1}^N \left( \frac{y_i - \hat{f}(\mathbf{x}_i)}{1 - \text{Tr}(S)/N} \right)^2, \quad \text{called the } \textit{Generalized Cross-Validation}.$$

## Link between GCV and AIC

- It turns out that there is a similarity between GCV and AIC, recall that:

$$AIC = \overline{\text{Err}} + \frac{2}{N} p \sigma^2.$$

- As  $\text{Tr}(S) = p$ , we get under a squared loss:

$$\begin{aligned} GCV &= \frac{1}{N} \sum_{i=1}^N (y_i - \hat{f}(\mathbf{x}_i))^2 \cdot \frac{1}{(1 - \frac{p}{N})^2} \\ \Rightarrow GCV &= \frac{RSS}{N} \cdot \frac{1}{(1 - \frac{p}{N})^2} \end{aligned}$$

- For  $|x| < 1$ , the geometric series is:

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}, \quad \text{differentiating:}$$

$$\sum_{n=0}^{\infty} n x^{n-1} = \frac{1}{(1-x)^2} \simeq 0 + 1 \cdot x^0 + 2x^1 = 1 + 2x \quad (\text{for } x \text{ small})$$

- Therefore:

$$GCV \simeq \frac{RSS}{N} \left( 1 + \frac{2p}{N} \right) = \frac{RSS}{N} + \frac{2p}{N} \cdot \frac{RSS}{N}$$

$\hookrightarrow$  The GCV approximation is equivalent to AIC if we replace the noise variance by  $RSS/N \equiv \hat{\sigma}^2$ .